$$\frac{}{\Delta \mid x : A \vdash x : A}\ \mathsf{hyp}$$

$$\frac{}{\Delta \mid \cdot \vdash () : \mathbf{1}}\ \mathbf{1}I \qquad \frac{\Delta \mid \Omega \vdash e : \mathbf{1} \quad \Delta \mid \Omega_L \Omega_R \vdash e' : C}{\Delta \mid \Omega_L \Omega \Omega_R \vdash \mathbf{match}\ e\ (() \Rightarrow e') : A}\ \mathbf{1}E$$

$$\frac{\Delta \mid \Omega\,(x : A) \vdash e : B}{\Delta \mid \Omega \vdash \lambda x.\, e : A \twoheadrightarrow B}\ {\twoheadrightarrow}I \qquad \frac{\Delta \mid \Omega \vdash e_1 : A \twoheadrightarrow B \quad \Delta \mid \Omega_A \vdash e_2 : A}{\Delta \mid \Omega\,\Omega_A \vdash e_1\, e_2 : B}\ {\twoheadrightarrow}E$$

$$\frac{\Delta \mid (x : A)\,\Omega \vdash e : B}{\Delta \mid \Omega \vdash \lambda x.\, e : A \rightarrowtail B}\ {\rightarrowtail}I \qquad \frac{\Delta \mid \Omega \vdash e_1 : A \rightarrowtail B \quad \Delta \mid \Omega_A \vdash e_2 : A}{\Delta \mid \Omega_A\,\Omega \vdash e_1\, e_2 : B}\ {\rightarrowtail}E$$

$$\frac{\Delta \mid \Omega_A \vdash e_1 : A \quad \Delta \mid \Omega_B \vdash e_2 : B}{\Delta \mid \Omega_A\,\Omega_B \vdash (e_1, e_2) : A \bullet B}\ {\bullet}I \qquad \frac{\Delta \mid \Omega \vdash e : A \bullet B \quad \Delta \mid \Omega_L\,(x : A)\,(y : B)\,\Omega_R \vdash e' : C}{\Delta \mid \Omega_L\,\Omega\,\Omega_R \vdash \mathbf{match}\ e\ ((x, y) \Rightarrow e') : C}\ {\bullet}E$$

$$\frac{\Delta, \alpha\ \mathsf{type} \mid \Omega \vdash e : A}{\Delta \mid \Omega \vdash e : \forall \alpha.\, A}\ \forall I \qquad \frac{\Delta \mid \Omega \vdash e : \forall \alpha. A(\alpha) \quad \Delta \vdash B\ \mathsf{type}}{\Delta \mid \Omega \vdash e : A(B)}\ \forall E$$